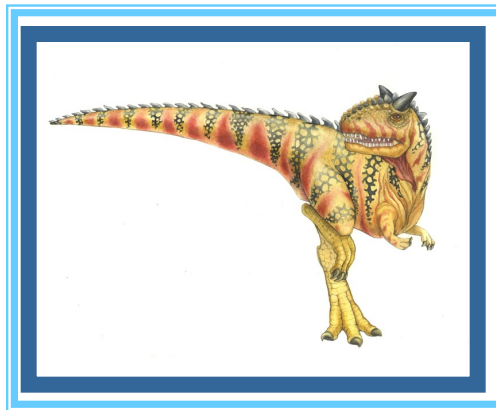
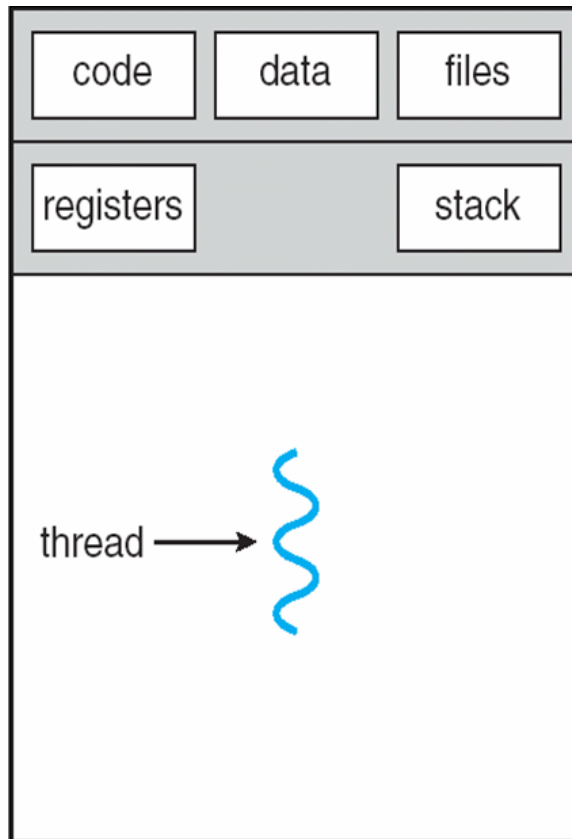


Chapter 4: Threads

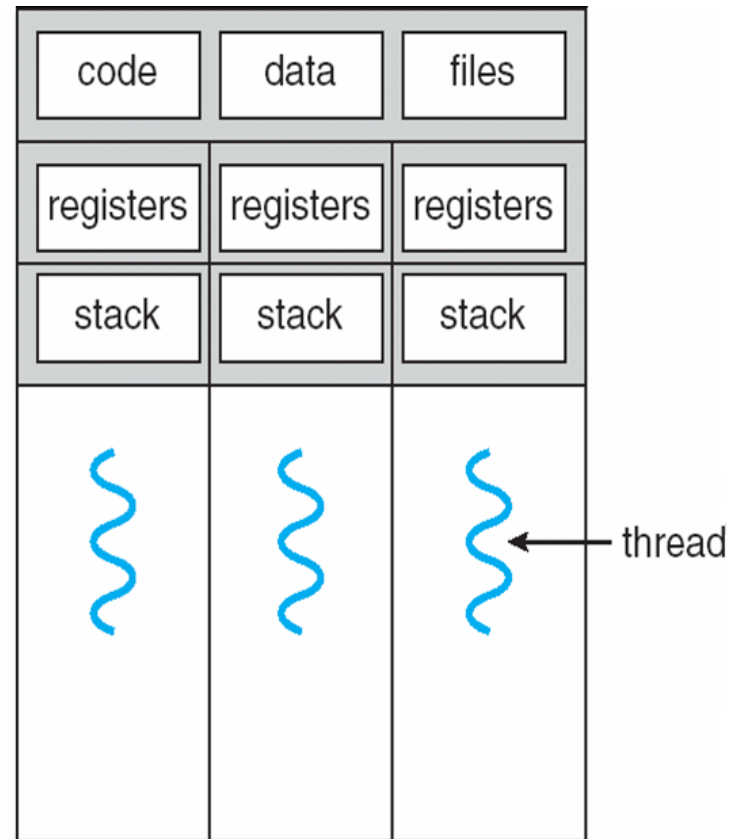




Single and Multithreaded Processes



single-threaded process



multithreaded process





Benefits

- Responsiveness
- Resource Sharing
- Economy
- Scalability





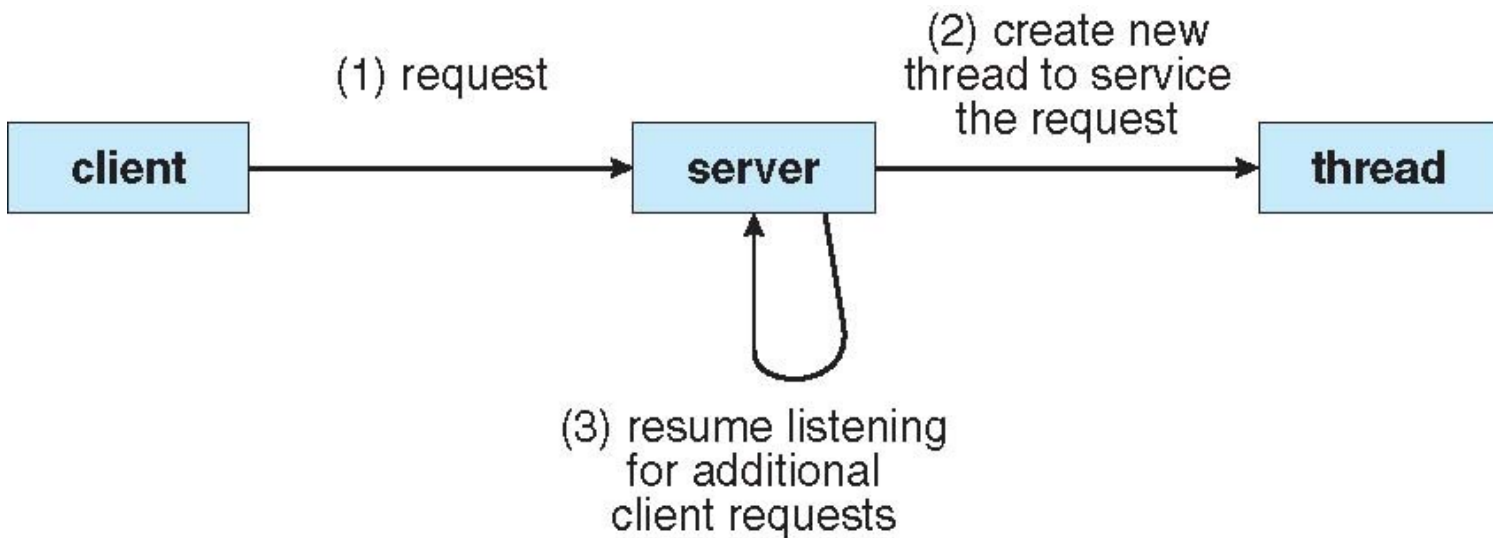
Multicore Programming

- Multicore systems putting pressure on programmers, challenges include
 - **Dividing activities**
 - **Balance**
 - **Data splitting**
 - **Data dependency**
 - **Testing and debugging**



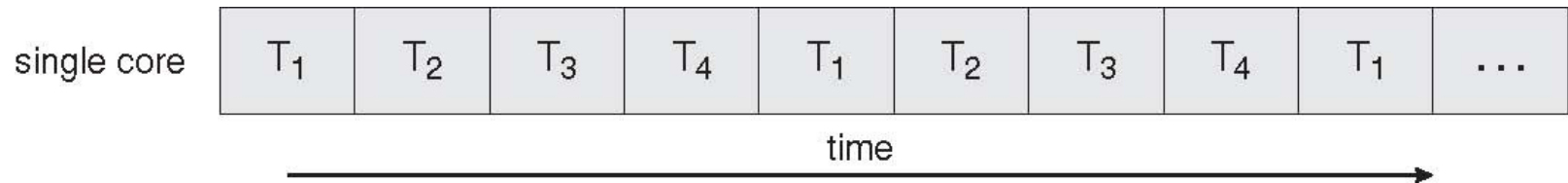


Multithreaded Server Architecture



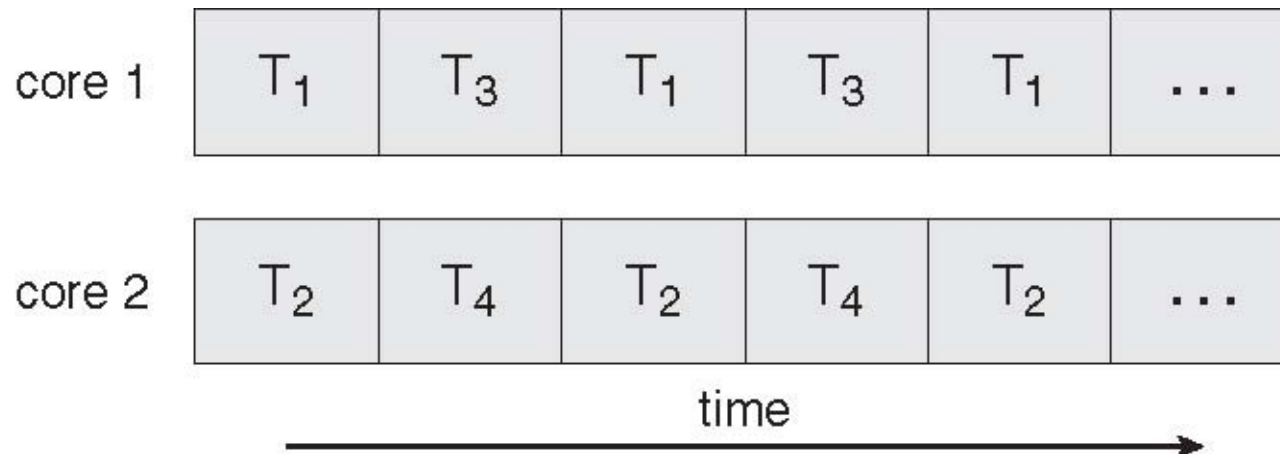


Concurrent Execution on a Single-core System





Parallel Execution on a Multicore System





User Threads

- Thread management done by user-level threads library
- Three primary thread libraries:
 - POSIX Pthreads
 - Win32 threads
 - Java threads





Kernel Threads

- Supported by the Kernel
- Examples
 - Windows XP/2000
 - Solaris
 - Linux
 - Tru64 UNIX
 - Mac OS X





Multithreading Models

- Many-to-One
- One-to-One
- Many-to-Many





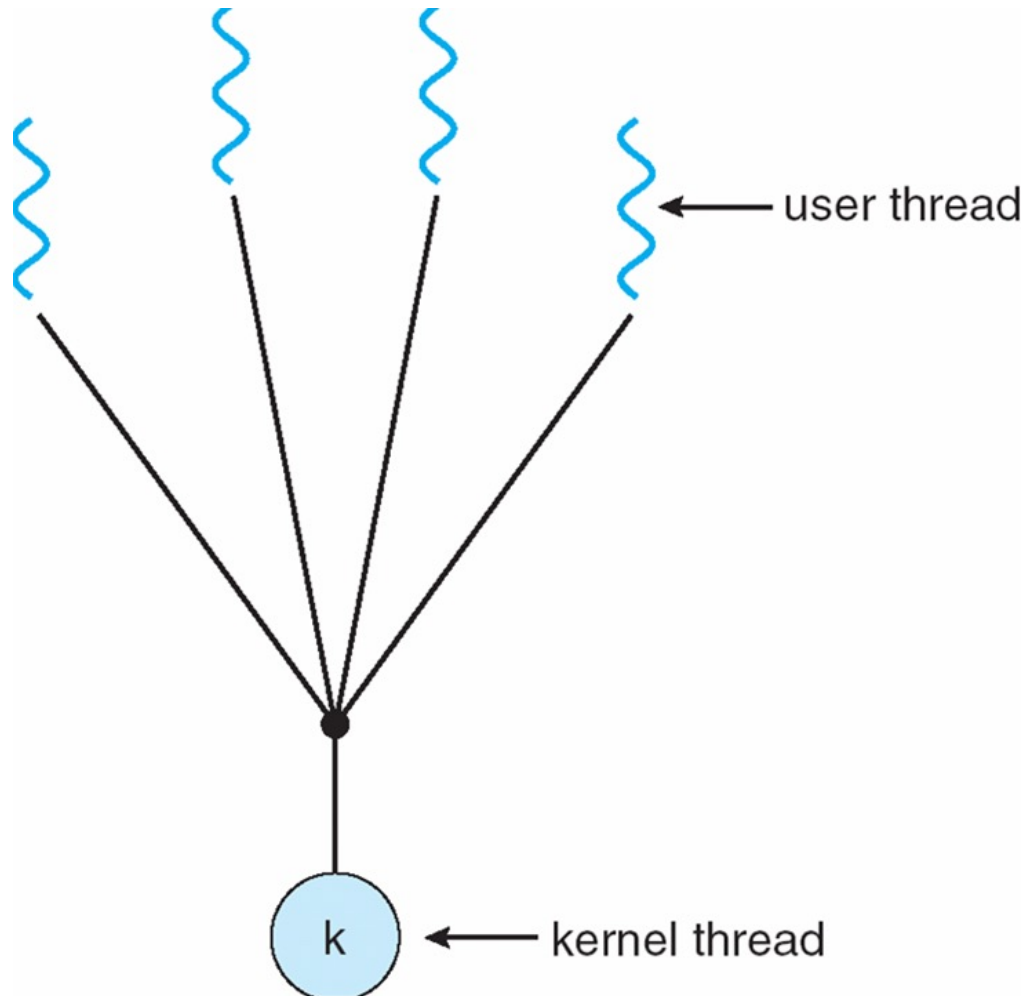
Many-to-One

- Many user-level threads mapped to single kernel thread
- Examples:
 - Solaris Green Threads
 - GNU Portable Threads





Many-to-One Model





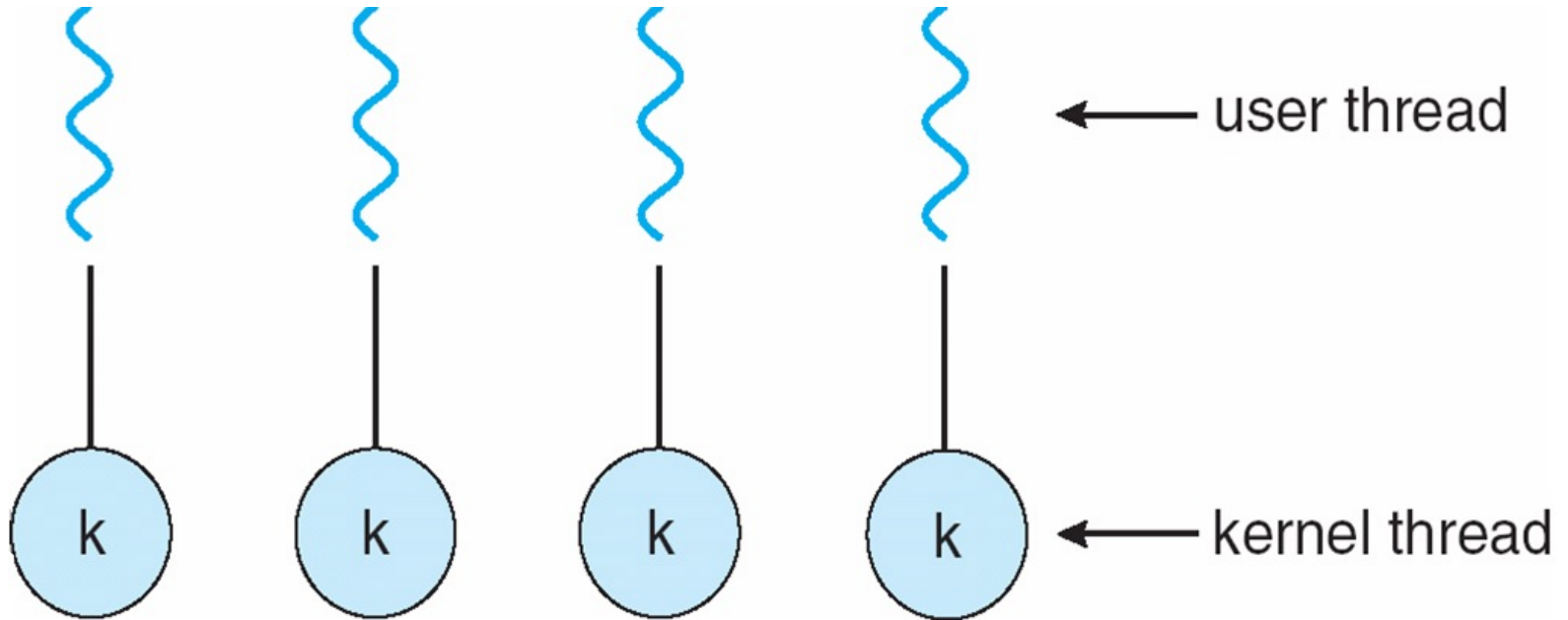
One-to-One

- Each user-level thread maps to kernel thread
- Examples
 - Windows NT/XP/2000
 - Linux
 - Solaris 9 and later





One-to-one Model





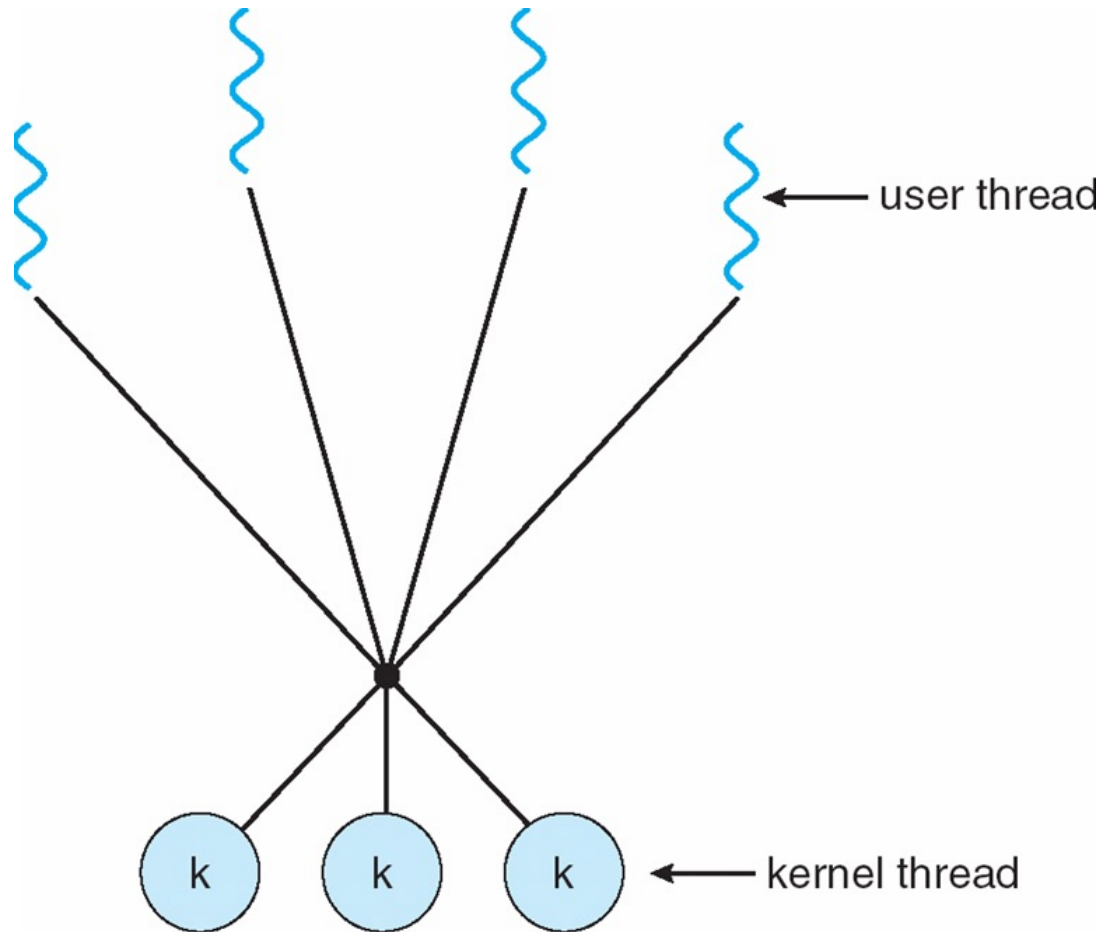
Many-to-Many Model

- Allows many user level threads to be mapped to many kernel threads
- Allows the operating system to create a sufficient number of kernel threads
- Solaris prior to version 9
- Windows NT/2000 with the *ThreadFiber* package





Many-to-Many Model





Two-level Model

- Similar to M:M, except that it allows a user thread to be **bound** to kernel thread
- Examples
 - IRIX
 - HP-UX
 - Tru64 UNIX
 - Solaris 8 and earlier





Two-level Model

